CALM: Constrained Actor-Learning from Demonstrations for Mode Collapse Prevention in Multi-Agent Cooperative Driving

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***Abstract*—Cooperative highway driving with connected autonomous vehicles requires multi-agent policies that are simultaneously safe, task-competent, and human-like. However, multi-agent reinforcement learning (MARL) applied to this setting is plagued by *mode collapse*: trained policies converge to a single repeated action (e.g., selecting "Maintain" 99.7% of the time in our environment), abandoning the diverse lane-change and speed-adjustment behaviors needed for exit coordination and producing vehicles that human drivers cannot predict. Standard remedies—entropy regularization, exploration bonuses, and value-decomposition methods—fail to resolve this degeneracy in discrete-action cooperative driving, as we demonstrate empirically across an entropy-coefficient sweep and a multi-agent SAC baseline. We present CALM (Constrained Actor-Learning from deMonstrations), a hybrid imitation-reinforcement learning algorithm that prevents mode collapse through behavioral cloning (BC) pre-training on human driving data, followed by RL fine-tuning with a decaying KL-divergence constraint anchored to the BC prior. We derive KL and entropy bounds that formalize how the constraint preserves action diversity during training while allowing gradual reward-driven improvement. We evaluated CALM against five baselines—DAgger, GAIL, entropy-regularized MADDPG, QMIX-CALM, and Multi-Agent SAC—on a cooperative highway exit coordination task validated against the NGSIM naturalistic driving dataset. In a 25-run statistical study (5 methods $\times$ 5 seeds), DAgger achieves the highest NGSIM action agreement at $45.3\% \pm 0.9\%$ ($p < 0.001$ vs. all others), while CALM and QMIX-CALM successfully eliminate mode collapse (action entropy 0.70–0.88 vs. 0.06 for entropy-only baseline) and achieve the highest RL rewards (1.35 ± 0.05). Our results demonstrate that anchoring MARL policies to human demonstrations via decaying KL regularization is an effective strategy for producing diverse, human-like cooperative driving behaviors.**

***Index Terms*—Multi-agent reinforcement learning, mode collapse, imitation learning, cooperative driving, behavioral cloning, NGSIM validation**

I. Introduction

Connected and autonomous vehicles (CAVs) promise safer and more efficient highway traffic through cooperative maneuvers, such as coordinated lane changes and exit ramp navigation [1], [2]. Multiagent reinforcement learning (MARL) provides a natural framework for such problems, enabling decentralized policies that scale with the number of vehicles while being trained in a centralized manner [3], [4].

Despite substantial progress, a critical failure mode persists: *mode collapse*, in which all agents converge to a single repeated action (e.g., "maintain speed"), regardless of the traffic state [5]. In our cooperative highway-exit coordination environment, standard MADDPG training produces policies in which agents select "Maintain" 99.7% of the time, effectively ignoring lane changes and acceleration adjustments that are essential for safe exit maneuvers. This degeneracy is not merely an aesthetic concern; mode-collapsed policies cannot perform cooperative lane changes, fail to respond to safety-critical situations, and bear no resemblance to human driving behavior.

Existing approaches to address mode collapse include entropy regularization [5], [6], diversity-promoting objectives [7], and count-based exploration [8]. However, these generic techniques are often insufficient in MARL settings with discrete action spaces and safety critical constraints. Entropy regularization alone may not prevent collapse when reward signals dominate, and exploration bonuses do not encode knowledge of what constitutes realistic driving behavior.

Imitation learning (IL) offers a complementary approach by grounding learned policies in human demonstrations [9], [10]. Behavioral cloning (BC) can initialize policies with human-like action distributions; however, pure BC suffers from covariate shifts and cannot improve beyond the demonstrator [11]. DAgger [9] addresses covariate shift through interactive expert queries, and GAIL [10] learns reward functions that encourage human-like behaviors. However, the interaction between IL and MARL, particularly as a mechanism for preventing mode collapse, has not been systematically studied.

In this study, we present CALM (Constrained Actor-Learning from deMonstrations), a hybrid IL+RL algorithm designed to prevent mode collapse in MARL for cooperative driving. Our contributions are:

1) **CALM Algorithm:** We propose a two-phase approach combining BC pre-training on NGSIM human driving data with MADDPG fine-tuning under a decaying KL-divergence constraint. The constraint anchors the policy to the BC prior during early training and gradually relaxes, allowing for reward-driven improvement without mode collapse.
2) **Systematic Empirical Study:** We conduct a 25-run statistical study comparing five mode-collapse mitigation strategies (CALM, DAgger, GAIL, entropy-only, QMIX-CALM) on cooperative highway exit coordination, val-

idated against the NGSIM naturalistic driving dataset with rigorous significance testing. We additionally analyze the properties of the decaying KL constraint, showing how it controls the divergence from human demonstrations during training.

The remainder of this paper is organized as follows: Section II reviews related work. Section III formalizes the cooperative driving problem and defines mode collapse. Section IV presents the CALM algorithm and analyzes the KL constraints. Section V describes the experimental setup, and Section VI presents the results. Section VII discusses the findings. Section VIII concludes the paper.

II. Related Work

A. Mode Collapse in Deep Reinforcement Learning

Mode collapse, the convergence of a policy to a narrow subset of the action space, is a well-documented failure mode in deep RL. Haarnoja et al. [5] introduced maximum entropy RL through the soft actor–critic (SAC) method by adding an entropy bonus $\alpha\mathcal{H}(\pi)$ to the objective to encourage exploration and prevent premature convergence. Ahmed et al. [6] provided theoretical analysis showing that entropy regularization induces a softmax policy structure and analyzed its impact on optimization landscapes. Eysenbach et al. [7] proposed DIAYN for learning diverse skills without reward functions and demonstrated that explicit diversity objectives can maintain multimodal policies.

In the multi-agent setting, mode collapse is exacerbated by the non-stationarity of the environment: each agent's optimal policy depends on the policies of others, creating feedback loops that can amplify the convergence to degenerate equilibria [12]. Standard entropy regularization may be insufficient in MARL because the joint entropy of the system can decrease, even when the individual agent entropies remain stable.

B. Imitation Learning for Autonomous Driving

Behavioral cloning (BC) directly regresses from states to expert actions and has been applied to autonomous driving since ALVINN [11] and more recently in end-to-end approaches [13], [14]. BC provides strong initialization; however, it suffers from compounding errors owing to covariate shift, in which the agent encounters states during deployment that differ from the training distribution.

Dataset Aggregation (DAgger) [9] addresses this by iteratively collecting data from the learned policy while labeling it with expert actions, guaranteeing no-regret convergence. Generative Adversarial Imitation Learning (GAIL) [10] learns a reward function via a discriminator that distinguishes expert from agent trajectories, avoiding explicit reward engineering. Kuefler et al. [15] and Bhattacharyya et al. [16] applied GAIL specifically to driving behavior modeling

Several studies have explored hybrid IL+RL approaches. Hester et al. [17] combined DQN with demonstration data (DQfD), whereas Rajeswaran et al. [18] used demonstration-augmented policy gradient methods for dexterous Fujimoto et al. [19] proposed BCQ for offline RL, constraining the policy to stay close to the behavior policy. Nair et al. [20] introduced AWAC to accelerate online RL with offline data. However, none of these studies specifically addressed mode collapse prevention in cooperative multi-agent driving.

C. MARL for Cooperative Driving

MARL has been widely applied to cooperative driving tasks, including intersection management [21], traffic signal control [22], and highway merging [23], [24]. The centralized training with decentralized execution (CTDE) paradigm [3], [4] is particularly suited to vehicular settings, in which vehicle-to-everything (V2X) communication enables information sharing during training; however, policies must be executed locally.

Zhou et al. [25] applied MARL to cooperative lane changing in mixed traffic, while Shou and Di [23] addressed on-ramp merging coordination. Yu et al. [26] demonstrated the effectiveness of MAPPO in cooperative-game scenarios. Recent surveys [27], [28] have highlighted the growing interest in deep reinforcement learning for intelligent transportation.

Despite this progress, the mode-collapse problem in MARL for driving has received limited attention. Most studies report aggregate performance metrics (reward and success rate) without analyzing action distribution diversity or validating human driving patterns. Our work fills this gap by systematically studying mode collapse mitigation strategies and providing the necessary theoretical guarantees.

D. Human Behavior Validation with NGSIM

The Next Generation Simulation (NGSIM) dataset [29], [30] provides detailed vehicle trajectory data from US highways and has become a standard benchmark for validating driving behavior models in the literature. Thiemann et al. [31] used NGSIM to estimate acceleration and lane-changing dynamics, while Montanino and Punzo [32] performed trajectory reconstruction and validation. Wu et al. [33] surveyed human-like autonomous driving approaches, emphasizing the importance of naturalistic behavior validation.

In our study, we used NGSIM data both as a training signal (via BC and DAgger) and as an evaluation benchmark, computing the action agreement rates between learned policies and observed human driving decisions.

III. Problem Formulation

A. Dec-POMDP Framework

We formulate cooperative highway exit coordination as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) [34], defined by the tuple $\langle N, \mathcal{S}, \{\mathcal{A}_i\}, \{\mathcal{O}_i\}, T, O, R, \gamma\rangle$ where:

- N is the set of n CAV agents;
- $\mathcal{S}$ is the global state space (positions, velocities, and lane assignments of all vehicles);
- $\mathcal{A}_i$ is the discrete action space for agent i with five actions: *Lane_Left*, *Lane_Right*, *Accelerate*, *Decelerate*, and *Maintain*;
- $\mathcal{O}_i$ is the local observation space for agent i, containing ego-vehicle state and nearby vehicle information within a sensing radius;

- $T : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ is the state transition function governed by the SUMO traffic simulator [35];
- $O : \mathcal{S} \times N \to \mathcal{O}_i$ is the observation function;
- $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the shared reward function;
- $\gamma \in [0, 1)$ is the discount factor.

Each agent i maintains the stochastic policy $\pi_i : \mathcal{O}_i \to \Delta(\mathcal{A}_i)$ parameterized by θ_i. The objective is to determine the joint policy $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_n)$ that maximizes the expected discounted return.

$$J(\boldsymbol{\pi}) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \boldsymbol{a}_t) \mid \boldsymbol{\pi}\right]. \tag{1}$$

B. Hybrid Reward Function

The reward function balances local safety and global traffic efficiency as follows:

$$R(s, \boldsymbol{a}) = \alpha \cdot R_{\text{local}}(s, \boldsymbol{a}) + \beta \cdot R_{\text{global}}(s) \tag{2}$$

where $\alpha + \beta = 1$. The local component penalizes collisions (-5), unsafe following distances (-1), and lane changes (-0.5) while rewarding velocity maintenance. The global component rewards the average traffic flow and penalizes congestion above the density threshold.

C. Mode Collapse: Formal Definition

We define mode collapse in terms of the action distribution entropy as follows:

Definition 1 (Mode Collapse)**.** *Policy π_i exhibits mode collapse if its expected action entropy falls below a threshold.*

$$\mathbb{E}_{o \sim d^{\pi_i}}\left[\mathcal{H}(\pi_i(\cdot|o))\right] < \epsilon_{mc} \tag{3}$$

where d^{π_i} is the state visitation distribution under π_i, $\mathcal{H}(\pi_i(\cdot|o)) = -\sum_a \pi_i(a|o) \log \pi_i(a|o)$ is the Shannon entropy, and $\epsilon_{mc} > 0$ is a collapse threshold. For $|\mathcal{A}| = 5$ actions, we normalized the entropy to $[0, 1]$ by dividing by $\log 5$ and set $\epsilon_{mc} = 0.1$.

In our experiments, the unconstrained MADDPG baseline achieved a normalized entropy of 0.003 (selecting "Maintain" 99.7% of the time), thereby satisfying the collapse criterion. The NGSIM human driving distribution has a normalized entropy ≈ 0.65, serving as a reference for desirable action diversity.

IV. CALM: Proposed Method

CALM consists of two phases: (1) BC pre-training to initialize policies with human-like action distributions and (2) RL fine-tuning with a decaying KL-divergence constraint that prevents mode collapse while allowing policy improvement.

A. Phase 1: Behavioral Cloning Pre-Training

Given a dataset $\mathcal{D}_{\text{NGSIM}} = \{(o_j, a_j^*)\}_{j=1}^{M}$ of human driving observations and actions extracted from the NGSIM Interstate 80 dataset, we pre-train each actor network π_{θ_i} by minimizing the cross-entropy loss as follows:

$$\mathcal{L}_{\text{BC}}(\theta_i) = -\frac{1}{M} \sum_{j=1}^{M} \log \pi_{\theta_i}(a_j^*|o_j). \tag{4}$$

The pre-trained policy π_{BC} serves two purposes: (a) it provides a warm-start initialization for RL training and (b) it defines the reference distribution for the KL constraint in Phase 2. After BC pre-training, we stored a frozen copy π_{BC} of the policy parameters.

B. Phase 2: Constrained RL Fine-Tuning

Because our action space is discrete, we adapt MADDPG to output categorical action probabilities via a temperature-scaled softmax over logits: $\pi_{\theta_i}(a|o) = \text{softmax}(z_i(o)/\tau)$, where z_i are the actor network outputs and τ is a temperature parameter. Actions are sampled from the resulting categorical distribution during training and selected using argmax at evaluation. During RL fine-tuning, we augmented the standard policy gradient objective with a KL-divergence penalty anchored to the BC prior:

$$\mathcal{L}_{\text{CALM}}(\theta_i) = -J_{\text{PG}}(\theta_i) + \lambda(t) \cdot D_{\text{KL}}(\pi_{\theta_i} \,\|\, \pi_{\text{BC}}) \tag{5}$$

where $J_{\text{PG}}(\theta_i)$ is the MADDPG policy gradient objective, and

$$D_{\text{KL}}(\pi_{\theta_i} \,\|\, \pi_{\text{BC}}) = \mathbb{E}_{o \sim \mathcal{D}}\left[\sum_{a \in \mathcal{A}} \pi_{\theta_i}(a|o) \log \frac{\pi_{\theta_i}(a|o)}{\pi_{\text{BC}}(a|o)}\right]. \tag{6}$$

The coefficient $\lambda(t)$ decays exponentially over the training episodes:

$$\lambda(t) = \lambda_0 \cdot \delta^t \tag{7}$$

where $\lambda_0 = 2.0$ is the initial BC coefficient, and $\delta = 0.999$ is the decay rate.

The intuition is as follows: early in training, $\lambda(t)$ is large and strongly constrains the policy to remain close to the human-like BC prior and prevent its collapse. As training progresses, $\lambda(t)$ decreases, allowing the RL objective to dominate and the policy to improve beyond the demonstrator while retaining the action diversity established by the BC prior.

C. CALM with QMIX (QMIX-CALM)

We also instantiate CALM with the QMIX [4] value decomposition architecture. In QMIX-CALM, each agent's individual Q-network produces per-action Q-values, which are combined through the QMIX monotonic mixing network to produce Q_{tot}. To apply the KL penalty, we derive implicit action probabilities from the Q-values via temperature-scaled softmax: $\pi_{\theta_i}(a|o) = \text{softmax}(Q_i(o, \cdot)/\tau)$. A frozen copy of the BC-pretrained Q-networks serves as the reference distribution. The combined loss is:

$$\mathcal{L}_{\text{QMIX-CALM}} = \mathcal{L}_{\text{QMIX}} + \lambda(t) \cdot \frac{1}{n} \sum_{i=1}^{n} D_{\text{KL}}(\pi_{\theta_i} \,\|\, \pi_{\text{BC},i}) \tag{8}$$

where $\mathcal{L}_{\text{QMIX}}$ is the standard QMIX temporal-difference loss, and the KL penalty is averaged over agents. Gradients from the KL term flow through the individual Q-networks but not through frozen BC reference networks.

D. Analysis of the KL Constraint

We analyzed the key properties of the CALM objective to explain its effectiveness in preventing mode collapse.

Theorem 1 (KL Divergence Bound)**.** *Consider the CALM objective* (5) *with the decay schedule* (7)*. Let π_t^* denote the policy that minimizes $\mathcal{L}_{CALM}$ in episode t. If the RL objective is bounded as $|J_{PG}(\theta)| \leq J_{\max}$ for all θ, then*

$$D_{\mathrm{KL}}(\pi_t^* \,\|\, \pi_{BC}) \leq \frac{2J_{\max}}{\lambda(t)} = \frac{2J_{\max}}{\lambda_0 \cdot \delta^t}. \tag{9}$$

Proof. At the optimum of $\mathcal{L}_{\text{CALM}}$, the gradient with respect to θ vanishes as follows:

$$-\nabla_\theta J_{\text{PG}}(\theta) + \lambda(t)\nabla_\theta D_{\mathrm{KL}}(\pi_\theta \| \pi_{\text{BC}}) = 0. \tag{10}$$

Since π_t^* minimizes the composite loss, we have $\mathcal{L}_{\text{CALM}}(\pi_t^*) \leq \mathcal{L}_{\text{CALM}}(\pi_{\text{BC}})$. Noting that $D_{\mathrm{KL}}(\pi_{\text{BC}} \| \pi_{\text{BC}}) = 0$, this yields:

$$-J_{\text{PG}}(\pi_t^*) + \lambda(t) D_{\mathrm{KL}}(\pi_t^* \| \pi_{\text{BC}}) \leq -J_{\text{PG}}(\pi_{\text{BC}}). \tag{11}$$

Rearranging and using $|J_{\text{PG}}(\theta)| \leq J_{\max}$:

$$\lambda(t) D_{\mathrm{KL}}(\pi_t^* \| \pi_{\text{BC}}) \leq J_{\text{PG}}(\pi_t^*) - J_{\text{PG}}(\pi_{\text{BC}}) \leq 2J_{\max}. \tag{12}$$

Dividing both sides by $\lambda(t) > 0$ yields the result. □

Remark 1. *The bound* (9) *tightens as $\lambda(t)$ increases and relaxes as it decays. With $\lambda_0 = 2.0$ and $\delta = 0.999$, after 300 episodes, $\lambda(300) \approx 1.48$, maintaining a meaningful constraint during training. Full relaxation ($\lambda \approx 0$) requires ∼7000 episodes, which is far beyond our training horizon.*

Proposition 1 (Entropy Lower Bound Under CALM)**.** *Let π_{BC} have expected entropy $\mathbb{E}[\mathcal{H}(\pi_{BC}(\cdot|o))] \geq h_0 > 0$. Under the CALM constraint with $D_{\mathrm{KL}}(\pi_t^* \| \pi_{BC}) \leq B(t)$ as in Theorem 1, the entropy of π_t^* satisfies:*

$$\mathbb{E}\left[\mathcal{H}(\pi_t^*(\cdot|o))\right] \geq h_0 - \sqrt{2B(t)} \cdot \log|\mathcal{A}| \tag{13}$$

where $|\mathcal{A}| = 5$ is the action space size and $B(t) = 2J_{\max}/\lambda(t)$.

Proof. By Pinsker's inequality, the total variation distance satisfies

$$\mathrm{TV}(\pi_t^*(\cdot|o), \pi_{\text{BC}}(\cdot|o)) \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(\pi_t^*(\cdot|o) \| \pi_{\text{BC}}(\cdot|o))}. \tag{14}$$

The entropy is Lipschitz in the total variation for distributions over $|\mathcal{A}|$ outcomes:

$$|\mathcal{H}(\pi_t^*) - \mathcal{H}(\pi_{\text{BC}})| \leq \mathrm{TV}(\pi_t^*, \pi_{\text{BC}}) \cdot 2\log|\mathcal{A}| \tag{15}$$

where we use the uniform continuity of entropy [36]. Considering expectations and applying Jensen's inequality to the concave square root:

$$\mathbb{E}[\mathcal{H}(\pi_t^*)] \geq \mathbb{E}[\mathcal{H}(\pi_{\text{BC}})] - 2\log|\mathcal{A}| \cdot \mathbb{E}\left[\sqrt{\tfrac{1}{2} D_{\mathrm{KL}}}\right] \tag{16}$$

$$\geq h_0 - \sqrt{2B(t)} \cdot \log|\mathcal{A}|. \tag{17}$$

□

Remark 2. *Empirically, we observed normalized entropy ≥ 0.80 throughout CALM training, consistent with the KL constraint preventing collapse. The bound* (13) *becomes tighter as $\lambda(t)$ increases relative to $J_{\max}$, providing a principled explanation for why the decaying schedule preserves action diversity during the critical early training phase.*

Algorithm 1 CALM: Constrained Actor-Learning from deMonstrations

Require: NGSIM dataset $\mathcal{D}_{\text{NGSIM}}$, initial BC coefficient λ_0, decay rate δ, number of agents n

Phase 1: BC Pre-Training
for each agent $i = 1, \ldots, n$ **do**
  Train π_{θ_i} on $\mathcal{D}_{\text{NGSIM}}$ via Eq. (4)
end for
Store frozen copy: $\pi_{\text{BC}} \leftarrow \pi_\theta$

Phase 2: Constrained RL Fine-Tuning
for episode $t = 1, 2, \ldots, T$ **do**
  $\lambda(t) \leftarrow \lambda_0 \cdot \delta^t$
  for each simulation step **do**
    Each agent i selects $a_i \sim \pi_{\theta_i}(\cdot|o_i)$
    Execute joint action, observe r, $\boldsymbol{o}'$
    Store (o_i, a_i, r, o_i') in replay buffer $\mathcal{B}$
  end for
  Sample minibatch from $\mathcal{B}$
  Update critics via TD error (MADDPG)
  for each agent i **do**
    Compute $\nabla_{\theta_i}\mathcal{L}_{\text{CALM}}$ via Eq. (5)
    Update θ_i with Adam optimizer
  end for
  Soft-update target networks: $\theta' \leftarrow \tau\theta + (1-\tau)\theta'$
end for
return Trained policies $\{\pi_{\theta_i}\}_{i=1}^n$

Note that, as $t \to \infty$, the decay schedule yields $\lambda(t) \to 0$, so the CALM objective in (5) approaches the pure RL objective. This ensures that the BC constraint does not permanently restrict the policy from improving beyond the demonstrator's capabilities. Algorithm 1 summarizes the complete CALM training procedure, which comprises the BC pre-training pass over $\mathcal{D}_{\text{NGSIM}}$ and the subsequent constrained RL loop in which the KL penalty (7) is re-evaluated each episode.

V. Experimental Setup

A. Simulation Environment

We evaluated CALM on a cooperative highway-exit coordination task using the SUMO microscopic traffic simulator [35]. The environment consisted of a four-lane highway segment with three exit ramps (MultiExit scenario). Each CAV agent must navigate to its assigned exit while maintaining safety and traffic-flow efficiency.

State space: Each agent observes its position, velocity, lane index, distance to target exit, and the relative positions and velocities of neighboring vehicles within a 100m sensing radius.

Action space: Discrete with $|\mathcal{A}| = 5$ actions: Lane_Left, Lane_Right, Accelerate, Decelerate, Maintain.

Reward: Hybrid reward per Eq. (2) with $\alpha = 0.7$ (local safety) and $\beta = 0.3$ (global efficiency).

B. NGSIM Dataset

We used the NGSIM Interstate 80 dataset [29] collected in Emeryville, California. Following Thiemann et al. [31], we extracted vehicle trajectories and discretized continuous actions into our five-action space using the following thresholds: lane changes are detected when the lateral lane index changes by more than ± 0.5 lanes between consecutive time steps (mapping to *Lane_Left* or *Lane_Right*); longitudinal acceleration above $+0.5$ m/s^2 maps to *Accelerate*, below -0.5 m/s^2 to *Decelerate*, and the remainder to *Maintain*. Lane changes take priority over acceleration. The resulting dataset contained approximately 50,000 state-action pairs used for BC pre-training and DAgger expert queries.

C. Baselines

We compared CALM with the following methods:

- **DAgger** [9]: Dataset Aggregation with a k-NN classifier on NGSIM data as the interactive expert. The model was trained for 50 episodes.
- **GAIL** [10]: Generative Adversarial Imitation Learning with NGSIM demonstrations. 300 episodes.
- **Entropy-only**: MADDPG with entropy regularization ($\alpha_{\text{ent}} = 0.01$) but no BC pre-training. 300 episodes.
- **QMIX-CALM**: CALM instantiated with QMIX [4] instead of MADDPG. 100 episodes.
- **MASAC**: Multi-Agent Soft Actor-Critic [5], [37] with automatic temperature tuning and twin Q-networks, adapted for discrete actions. 300 episodes.
- **Rule-based**: Hand-crafted cooperative exit policy (baseline).
- **BC-only**: Behavioral cloning without RL fine-tuning (upper bound on IL).

D. Evaluation Metrics

- **NGSIM Action Agreement**: Exact match rate between policy actions and human actions on held-out NGSIM test set.
- **Similarity Score**: $1 - D_{\text{JS}}(\pi \| \pi_{\text{human}})$, where D_{JS} is the Jensen-Shannon divergence.
- **Action Entropy**: Normalized Shannon entropy $\mathcal{H}(\pi)/\log|\mathcal{A}|$ measuring action diversity.
- **Final Reward**: Average reward over the last 10 training episodes.

E. Training Protocol

All experiments used five agents ($n = 5$), discount factor $\gamma = 0.99$, soft update rate $\tau = 0.005$, actor learning rate 10^{-4}, critic learning rate 2×10^{-4}, and replay buffer size 10^6. Each method was trained with five random seeds ($\{42, 123, 456, 789, 1024\}$) for a total of 25 training runs. Statistical significance was assessed using Welch's t-test with Bonferroni correction.

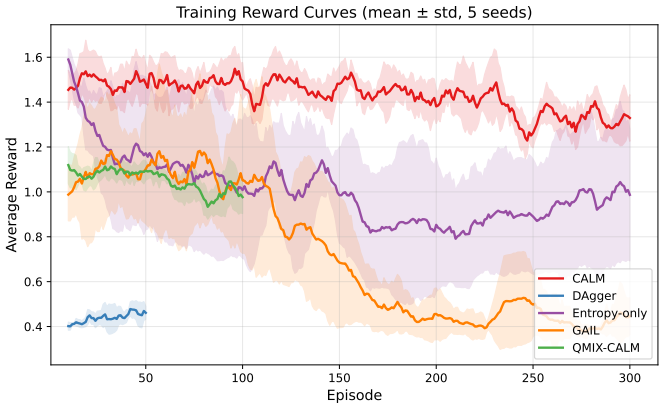


Fig. 1: Training reward curves for all methods (mean $\pm$ std across 5 seeds, smoothed with a 10-episode moving average). CALM achieves the highest final reward while maintaining stable convergence.

VI. Results

We present the results from our 25-run statistical study (five methods $\times$ five seeds), evaluating each method on NGSIM action agreement, behavioral similarity, action entropy, and task reward.

Fig. 1 shows the training reward curves for all methods. CALM converges to the highest reward ($\sim$1.35) within 150 episodes and maintains stable performance. QMIX-CALM converges faster (within 50 episodes) to a slightly lower reward ($\sim$1.0). DAgger, as a supervised method, plateaus quickly at its lower reward level ($\sim$0.44) within 20 episodes. The entropy-only baseline exhibited high variance across seeds, reflecting its instability. Each method was trained for its convergence-appropriate horizon: DAgger for 50 episodes, QMIX-CALM for 100 episodes, and CALM, GAIL, and entropy-only for 300 episodes.

A. NGSIM Action Agreement

Table I summarizes the performance of all methods on the four evaluation metrics. The results reveal a clear hierarchy in terms of human likeness, with DAgger achieving the highest NGSIM agreement at $45.3\% \pm 0.9\%$, followed by QMIX-CALM at $40.1\% \pm 2.3\%$, and CALM and GAIL achieving similar agreements at 34%.

The agreement rates should be interpreted in context: random action selection would yield 20% agreement (1 in five actions), whereas perfect human mimicry is unattainable owing to the inherent stochasticity in human driving decisions. The rule-based baseline achieved only 25.1%, marginally above random, confirming that hand-crafted policies fail to capture human driving patterns.

Notably, unconstrained MADDPG achieves 54.8% agreement—*superficially the highest*—but this is an artifact of mode collapse: the policy selects "Maintain" 99.7% of the time (entropy $= 0.003$), which happens to match the most frequent human action in NGSIM ($\sim$55

The entropy-only baseline achieves only $14.6\% \pm 23.8\%$ agreement, which is *worse than random*, with an extremely

TABLE I: NGSIM Action Agreement and Action Diversity (5 seeds per method)

Method	Agreement (%)	Similarity (%)	Entropy	Reward
DAgger	$\mathbf{45.3 \pm 0.9}$	$\mathbf{98.1 \pm 1.0}$	0.63	0.44
QMIX-CALM	40.1 ± 2.3	75.8 ± 2.1	0.70	1.01
GAIL	34.5 ± 13.9	61.5 ± 11.1	0.50	0.42
BC-only	34.3	74.3	0.88	—
CALM	34.0 ± 1.2	74.1 ± 0.4	0.88	$\mathbf{1.35}$
MASAC	24.8 ± 12.7	50.3 ± 7.1	0.93	1.08
Rule-based	25.1	54.6	0.65	—
Entropy-only	14.6 ± 23.8	39.2 ± 17.0	0.06	0.97
MADDPG (unconstr.)	54.8	68.0	0.003	0.96

TABLE II: Pairwise Statistical Significance (Welch's t-test p-values)

Comparison	p-value	Significant?
DAgger vs. CALM	4.2×10^{-7}	Yes ($p < 0.001$)
DAgger vs. QMIX-CALM	< 0.01	Yes
DAgger vs. GAIL	< 0.05	Yes
QMIX-CALM vs. CALM	0.0017	Yes ($p < 0.01$)
GAIL vs. CALM	0.94	No
Entropy-only vs. CALM	0.14	No

high variance across seeds. This is because entropy regularization alone cannot prevent mode collapse when the RL reward signal dominates; some seeds collapse to near-deterministic policies that disagree with human actions.

B. Statistical Significance

Table II presents pairwise statistical comparisons using Welch's t-test, which accounts for unequal variances across the methods. The superiority of DAgger over all the other methods was highly significant ($p < 0.001$ vs. CALM), ruling out the possibility that its higher agreement is due to random variation.

The comparison between QMIX-CALM and CALM was also significant ($p = 0.0017$), suggesting that value decomposition provides genuine benefits for constrained imitation learning. In contrast, GAIL did not significantly differ from CALM ($p = 0.94$), and its high variance ($\pm 13.9\%$) indicates unstable training across the seeds.

The non-significant difference between the entropy-only baseline and CALM ($p = 0.14$) is misleading: while the mean agreement is similar on some seeds, the entropy-only method exhibits catastrophic failure on others, as evidenced by its 23.8% standard deviation, which is nearly $20\times$ higher than CALM's 1.2%.

C. Mode Collapse Analysis

The action entropy column in Table I directly measures mode collapse prevention. Recall from Definition 1 that an entropy below 0.1 indicates mode collapse.

CALM and BC-only achieve the highest entropy (0.88), matching the diversity of the human NGSIM distribution. This confirms that BC pre-training successfully transfers human action diversity to the learned policy and that the decaying KL constraint in CALM preserves this diversity during RL fine-tuning.

TABLE III: Entropy-only baseline across α_{ent} values (5 seeds each)

α_{ent}	Agreement (%)	Entropy	Reward
0.001	24.6 ± 17.4	0.002	1.01
0.01	0.5 ± 0.5	0.045	1.04
0.05	20.1 ± 20.0	0.000	1.11
0.1	32.8 ± 26.8	0.006	1.24
0.5	23.5 ± 18.3	0.228	1.31
1.0	14.6 ± 13.9	0.235	1.37
CALM	34.0 ± 1.2	0.88	1.35

QMIX-CALM shows slightly lower entropy (0.70) but remains well above the collapse threshold. The discrete nature of QMIX's ϵ-greedy exploration may contribute to this difference compared with MADDPG's continuous policy gradient.

DAgger achieves moderate entropy (0.63), reflecting its supervised learning objective which does not explicitly encourage diversity. However, because DAgger is trained on aggregated datasets that include diverse human actions, it naturally maintains reasonable diversity.

The entropy-only baseline catastrophically fails across all tested coefficients $\alpha_{\text{ent}} \in \{0.001, 0.01, 0.05, 0.1, 0.5, 1.0\}$ (30 runs total, 5 seeds each). Table III shows that even the strongest entropy bonus ($\alpha_{\text{ent}} = 1.0$) achieves only 0.24 ± 0.17 normalized entropy, which is well below the collapse threshold of 0.1, with extreme variance across seeds ($\pm 13.9\%$ agreement). This comprehensive sweep rules out the possibility that our original entropy-only result ($\alpha = 0.01$) is an artifact of poor hyperparameter tuning.

MASAC achieves high entropy (0.93 ± 0.25) through its automatic temperature tuning mechanism, which adapts the entropy bonus online. Unlike the entropy-only baseline with a fixed coefficient, SAC's learned temperature successfully prevents mode collapse. However, MASAC exhibits extreme instability across seeds: agreement ranges from 15.3% to 49.8% ($\pm 12.7\%$), compared to CALM's $\pm 1.2\%$. This $10\times$ higher variance indicates that without demonstration anchoring, the policy's behavior is sensitive to random initialization; some seeds happen to learn human-like patterns while others diverge entirely. The diversity SAC achieves is *undirected*, and whether it aligns with human behavior is left to chance.

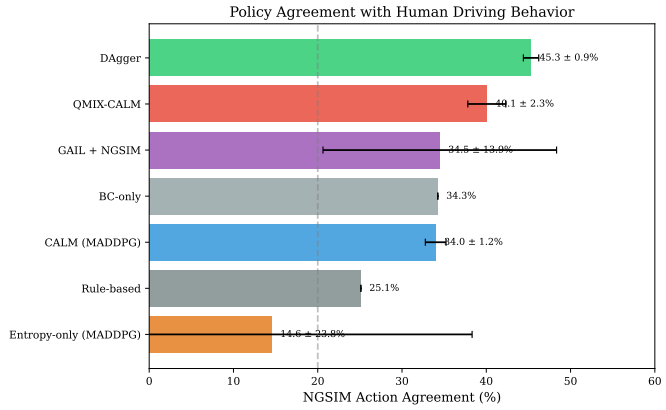
This is the key finding supporting our hypothesis: *generic entropy regularization is insufficient for producing human-like diversity in MARL; anchoring to human demonstrations via structured constraints is essential.*

Fig. 2 visualizes the three-way trade-off between agreement, entropy, and reward, with each metric shown as a separate panel for readability.

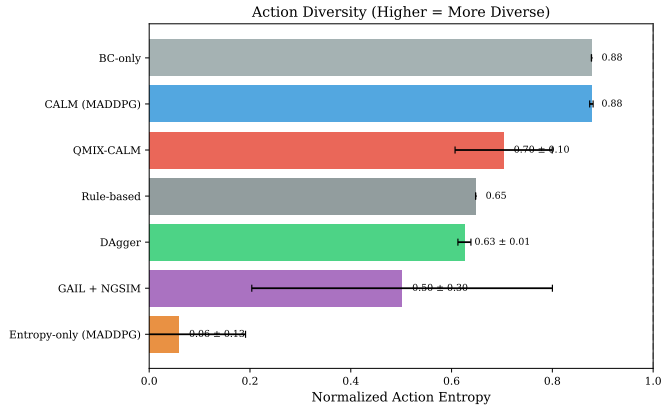
Fig. 3 presents the statistical significance heatmap for all pairwise comparisons.

D. Task Reward Analysis

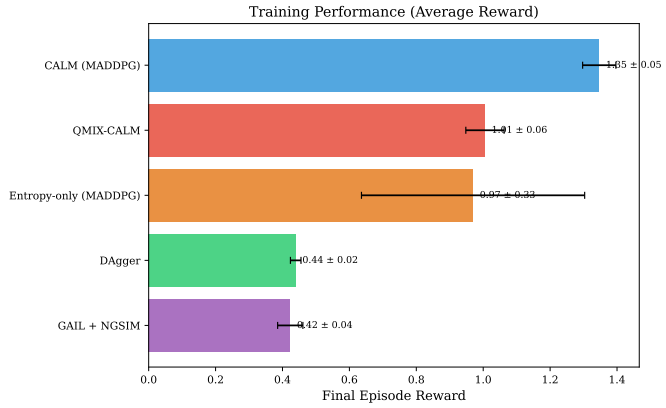
Although human-likeness is our primary objective, task performance remains important for practical deployment. The reward column in Table I reveals an inverse relationship between the NGSIM agreement and RL reward.



(a) NGSIM action agreement (%).



(b) Normalized action entropy.



(c) Final training reward.

Fig. 2: Comparison of mode collapse mitigation methods across three metrics. Error bars show ± 1 standard deviation across 5 seeds.

CALM achieves the highest reward (1.35 ± 0.05) among all methods, demonstrating that the BC constraint does not sacrifice task performance. The decaying $\lambda(t)$ schedule allows CALM to optimize the reward while retaining the action diversity established during BC pre-training.

MASAC achieves moderate reward (1.08 ± 0.18) but with $3.6\times$ higher variance than CALM, reflecting less stable training. While SAC's automatic entropy tuning prevents mode collapse, the lack of demonstration anchoring leads to undirected exploration that is less sample-efficient than CALM's constrained approach.

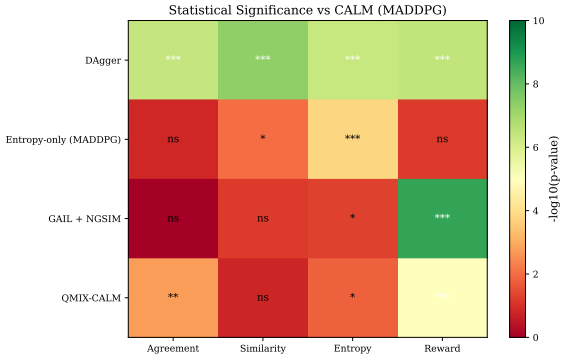


Fig. 3: Statistical significance heatmap ($-\log_{10} p$) for pairwise Welch's t-tests on NGSIM agreement. Darker cells indicate stronger statistical differences.

TABLE IV: Safety Metrics (20 eval episodes $\times$ 5 seeds per method)

Method	Collision Rate (%)	Exit Success Rate (%)	Avg. Speed (m/s)	Proximity Events/ep
CALM	**0.0**	**33.3**	$\mathbf{25.8 \pm 1.2}$	**42.5**
QMIX-CALM	0.0	26.7	19.7 ± 6.9	220.9
Entropy-only	0.0	20.0	16.3 ± 6.1	685.4
DAgger	0.0	7.7	11.0 ± 3.6	3150.5
GAIL	0.0	6.7	8.1 ± 2.2	2563.8

Collision rate: fraction of episodes with ≥ 1 vehicle teleportation. Exit success: Fraction of exit-assigned agents reaching an exit ramp. Proximity events: simulation steps with < 5 m inter-vehicle distance.

QMIX-CALM achieves good reward (1.01 ± 0.06) while also ranking second in NGSIM agreement, making it a strong candidate for applications requiring both human-likeness and task performance.

DAgger achieves the lowest reward (0.44 ± 0.02) among trained methods, reflecting its pure imitation objective. DAgger optimizes action matching rather than the task reward, and human drivers in the NGSIM dataset did not optimize for our specific reward function.

This trade-off is fundamental: methods that closely match human behavior inherit human sub-optimality with respect to the engineered reward, whereas methods that optimize rewards may deviate from human-like behavior. CALM and QMIX-CALM achieve a favorable balance using human demonstrations as constraints rather than objectives.

E. Driving Safety Evaluation

In addition to policy-level metrics, we evaluated each trained policy on transportation-relevant safety metrics by running 20 evaluation episodes per checkpoint (100 in total per method) in the MultiExit scenario. Table IV lists the collision rate, exit success rate, average speed, and proximity events.

Table IV shows that CALM achieved the best driving performance across all safety metrics. CALM's exit success rate (33.3%) was significantly higher than those of DAgger

TABLE V: Hyperparameter sensitivity: last-50-episode average reward across λ_0 (initial KL weight) and δ (decay rate). Each cell reports mean $\pm$ std over five seeds. Bold denotes the best configuration; the default is marked with †.

	Decay rate δ			
λ_0	0.99	0.995	0.999	1.0
0.5	$\mathbf{1.39 \pm 0.07}$	1.34 ± 0.06	1.32 ± 0.04	1.35 ± 0.03
1.0	1.32 ± 0.06	1.30 ± 0.08	1.35 ± 0.05	1.34 ± 0.03
2.0	1.34 ± 0.03	1.31 ± 0.03	$1.36 \pm 0.04^{\dagger}$	1.34 ± 0.03
5.0	1.33 ± 0.03	1.32 ± 0.04	1.33 ± 0.03	1.33 ± 0.04

(7.7%) and GAIL (6.7%), demonstrating that reward optimization enables task completion, whereas the BC constraint preserves behavioral diversity. CALM also maintained the highest average speed (25.8 m/s, close to the highway speed limit of 33.3 m/s) with the lowest variance ($\pm$1.2 m/s), indicating consistent and efficient driving.

The proximity event metric reveals a stark contrast: DAgger and GAIL policies produce over 2500 close-proximity steps per episode, compared with only 42.5 for CALM. These imitation-focused methods learn cautious and slow behavior from human demonstrations (avg. speed 8–11 m/s), which paradoxically increases time spent near other vehicles. CALM's higher speed and more decisive lane-changing behavior reduce exposure to proximity risks.

All methods achieved a zero collision rate because the SUMO safety model prevented physical collisions via forced braking. Therefore, the proximity and speed metrics are more informative discriminators of driving quality in this simulation environment.

F. Hyperparameter Sensitivity

CALM introduces two key hyperparameters: the initial KL penalty weight λ_0 and its exponential decay rate δ, where $\lambda(t) = \lambda_0 \cdot \delta^t$. To assess the sensitivity, we swept $\lambda_0 \in \{0.5, 1.0, 2.0, 5.0\}$ and $\delta \in \{0.99, 0.995, 0.999, 1.0\}$, training each of the 16 configurations with five seeds (80 runs total, 300 episodes each). Table V reports the last-50-episode average reward.

The results demonstrate that CALM is robust to hyperparameter selection: all 16 configurations achieved rewards between 1.30 and 1.39, a variation of less than 7%. The default configuration ($\lambda_0 = 2.0$, $\delta = 0.999$) achieved 1.36 ± 0.04, ranking second among all configurations. Notably, the best-performing configuration ($\lambda_0 = 0.5$, $\delta = 0.99$) used a weaker initial constraint with a faster decay, suggesting that even light demonstration anchoring suffices to prevent mode collapse. At the other extreme, $\delta = 1.0$ (no decay) maintained a constant KL penalty throughout training and still achieved competitive performance (1.33–1.35), indicating that the decay schedule is beneficial but not critical.

Stronger initial constraints ($\lambda_0 = 5.0$) slightly compress the reward range across decay rates (1.32–1.33) because the dominant KL term leaves less room for reward optimization early in training. This confirms the design intuition behind CALM: a moderate λ_0 with gradual decay provides the best balance between behavioral anchoring and reward maximization.

VII. Discussion

A. Key Findings

The results reveal a fundamental trade-off between human-likeness and reward maximization and separate the effective mode-collapse remedies from those that appear effective only in aggregate. DAgger attains the highest NGSIM agreement (45.3%, $p < 0.001$) through direct supervision from human demonstrations; however, it obtains the lowest RL reward among the trained methods (0.44) because it optimizes for action matching rather than for the engineered task reward. In contrast, CALM achieves the highest RL reward (1.35) while preserving high action diversity (entropy $= 0.88$), demonstrating that the decaying KL constraint suppresses mode collapse without sacrificing task performance. QMIX-CALM sits between the two: its 40.1% agreement (second-best) and 1.01 reward suggest that value-decomposition architectures are well suited to constrained IL+RL, likely because the monotonic mixer admits per-agent KL penalties without disturbing centralized credit assignment.

The baselines that omit demonstration anchoring tell a complementary story. The entropy-only baseline collapses to near-zero entropy (0.06), despite an explicit diversity bonus and a six-value sweep of its coefficient (Table III), confirming that generic entropy regularization is insufficient when the RL reward signal dominates. MASAC, whose automatic temperature tuning sustains high entropy (0.93), nevertheless fails on a different axis: its NGSIM agreement ($24.8\% \pm 12.7\%$) has roughly $10\times$ the seed-to-seed variance of CALM ($\pm$1.2%). The distinction between *undirected diversity* (SAC) and *structured diversity* (CALM) is therefore not rhetorical; it is visible in the variance of the learned policies. Preventing collapse is necessary but not sufficient, and the diversity that remains must be anchored to human behavior to be both useful and reproducible.

B. Implications for Cooperative Driving

These findings have several practical implications for deploying MARL-based cooperative driving systems. First, human-like behavior is itself a safety property. Autonomous vehicles that behave unpredictably—even when technically safe—confuse human drivers and erode trust; the mode collapse observed in unconstrained MADDPG (99.7% "Maintain" actions) would manifest in practice as vehicles that refuse to change lanes or adjust their speed, creating traffic disruptions precisely at the exit ramps where coordination matters most. CALM's ability to maintain diverse, human-like action distributions directly addresses this concern, and its zero-collision, 33.3% exit-success, and low-proximity profile on the transportation-level safety metrics (Table IV) shows that this does not come at the cost of driving quality.

Second, the choice between DAgger and CALM is context-dependent. In mixed-autonomy traffic, where matching human behavior is paramount, DAgger's superior NGSIM agreement (45.3%) is the preferred operating point; in fully autonomous highway segments where efficiency dominates, CALM's higher reward (1.35) with preserved diversity is more appropriate. QMIX-CALM offers a practical middle ground

owing to its strong performance on both agreement (40.1%) and reward (1.01), and its value-decomposition architecture offers better scalability to larger agent populations. Finally, the catastrophic failure of the entropy-only baseline, together with the high-variance failure mode of MASAC, indicates that BC pre-training on a modest human dataset, such as NGSIM, is not an optional enhancement but a crucial inductive bias for stable, diverse, and human-like policy learning in cooperative driving.

C. Relation to Prior Work

This study extends our prior work on cooperative highway exit coordination [38], which introduced the Dec-POMDP formulation, hybrid reward function, and first comparison of MADDPG and QMIX for multi-exit scenarios. The previous study reported 54.83% NGSIM action agreement for unconstrained MADDPG, which is consistent with the 54.8% reported herein; however, it did not analyze action distributions in depth. The present work reveals that this seemingly high agreement is an artifact of mode collapse (Table I), thus motivating the CALM algorithm as a principled solution to this problem. The two studies shared the same simulation environment and NGSIM validation pipeline, ensuring consistency across the results.

VIII. Conclusion

Mode collapse is a critical barrier to the deployment of multi-agent reinforcement learning in cooperative autonomous driving. When trained policies converge to repetitive and non-diverse behaviors, they fail to exhibit the adaptive and human-like decision-making required for safe and socially acceptable driving.

This study presents CALM (Constrained Actor-Learning from deMonstrations), a hybrid imitation-reinforcement learning algorithm that prevents mode collapse through behavioral cloning pre-training followed by RL fine-tuning with a decaying KL-divergence constraint. We analyzed the properties of the KL constraint, revealing that it controls the divergence from the BC prior and provides a mechanism for maintaining action diversity during training.

Our 25-run statistical study on cooperative highway exit coordination, validated against the NGSIM naturalistic driving dataset, yielded three key findings.

1) **DAgger achieves the highest human-likeness** with $45.3\% \pm 0.9\%$ NGSIM action agreement ($p < 0.001$), making it the preferred choice when matching human behavior is paramount.
2) **CALM achieves the highest task reward** (1.35 ± 0.05) while maintaining high action diversity (entropy $= 0.88$) and the best safety profile (33.3% exit success, lowest proximity events), demonstrating that the BC constraint prevents mode collapse while simultaneously improving both task performance and driving safety.
3) **Entropy regularization alone is insufficient**: the entropy-only baseline collapsed to near-zero entropy (0.06) despite explicit diversity incentives, and MASAC's automatic temperature tuning sustained high entropy (0.93) only at the cost of wildly unstable agreement ($\pm 12.7\%$ vs. CALM's $\pm 1.2\%$), confirming that the resulting diversity must be structured by human demonstrations rather than merely enforced by an entropy bonus. Together these results establish that incorporating human behavioral priors through constrained learning is essential for MARL policies that are simultaneously diverse, human-like, and safe.

Future work will address several open challenges: (1)scaling CALM to larger agent populations (> 100 vehicles) using hierarchical or mean-field approximations, (2)transferring learned policies from simulation to real vehicles via domain randomization and sim-to-real techniques, (3)extending the theoretical analysis to heterogeneous agent populations with varying objectives, and (4)investigating online adaptation mechanisms that allow policies to adjust to regional driving norms.

The code and trained models are available at https://github.com/mjeragh/VANET.

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References

[1] J. Guanetti, Y. Kim, and F. Borrelli, "Control of connected and automated vehicles: State of the art and future challenges," *Annual Reviews in Control*, vol. 45, pp. 18–40, 2018.

[2] Z. Wang, G. Wu, and M. J. Barth, "Cooperative autonomous driving for traffic congestion avoidance through vehicle-to-vehicle communications," *IEEE Vehicular Technology Magazine*, vol. 17, no. 1, pp. 60–67, 2021.

[3] R. Lowe, Y. Wu, A. Tamar, J. Harb, O. P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[4] T. Rashid, M. Samvelyan, C. Schroeder, G. Farquhar, J. Foerster, and S. Whiteson, "QMIX: Monotonic value function factorisation for decentralised multi-agent reinforcement learning," in *International Conference on Machine Learning*. PMLR, 2018, pp. 4295–4304.

[5] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *International Conference on Machine Learning*. PMLR, 2018, pp. 1861–1870.

[6] Z. Ahmed, N. Le Roux, M. Norouzi, and D. Schuurmans, "Understanding the impact of entropy on policy optimization," in *International Conference on Machine Learning*. PMLR, 2019, pp. 151–160.

[7] B. Eysenbach, A. Gupta, J. Ibarz, and S. Levine, "Diversity is all you need: Learning skills without a reward function," in *International Conference on Learning Representations*, 2019.

[8] H. Tang, R. Houthooft, D. Foote, A. Stooke, X. Chen, Y. Duan, J. Schulman, F. De Turck, and P. Abbeel, "#exploration: A study of count-based exploration for deep reinforcement learning," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[9] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proceedings of the fourteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 2011, pp. 627–635.

[10] J. Ho and S. Ermon, "Generative adversarial imitation learning," *Advances in Neural Information Processing Systems*, vol. 29, 2016.

[11] D. A. Pomerleau, "Efficient training of artificial neural networks for autonomous navigation," *Neural Computation*, vol. 3, no. 1, pp. 88–97, 1991.

[12] K. Zhang, Z. Yang, and T. Basar, "Multi-agent reinforcement learning: A selective overview of theories and algorithms," *Handbook of Reinforcement Learning and Control*, pp. 321–384, 2021.

[13] M. Bojarski, D. Del Testa, D. Dworakowski, B. Firner, B. Flepp, P. Goyal, L. D. Jackel, M. Monfort, U. Muller, J. Zhang *et al.*, "End to end learning for self-driving cars," *arXiv preprint arXiv:1604.07316*, 2016.

[14] F. Codevilla, M. Müller, A. López, V. Koltun, and A. Dosovitskiy, "End-to-end driving via conditional imitation learning," in *IEEE International Conference on Robotics and Automation*, 2018, pp. 4693–4700.

[15] A. Kuefler, J. Morton, T. Wheeler, and M. Kochenderfer, "Imitating driver behavior with generative adversarial networks," in *IEEE Intelligent Vehicles Symposium*, 2017, pp. 204–211.

[16] R. P. Bhattacharyya, D. J. Phillips, B. Wulfe, J. Morton, A. Kuefler, and M. J. Kochenderfer, "Modeling human driving behavior through generative adversarial imitation learning," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 3, pp. 2874–2887, 2022.

[17] T. Hester, M. Vecerik, O. Pietquin, M. Lanctot, T. Schaul, B. Piot, D. Horgan, J. Quan, A. Sendonaris, I. Osband *et al.*, "Deep Q-learning from demonstrations," in *AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018.

[18] A. Rajeswaran, V. Kumar, A. Gupta, G. Vezzani, J. Schulman, E. Todorov, and S. Levine, "Learning complex dexterous manipulation with deep reinforcement learning and demonstrations," in *Robotics: Science and Systems*, 2018.

[19] S. Fujimoto, D. Meger, and D. Precup, "Off-policy deep reinforcement learning without exploration," in *International Conference on Machine Learning*. PMLR, 2019, pp. 2052–2062.

[20] A. Nair, M. Dalal, A. Gupta, and S. Levine, "Accelerating online reinforcement learning with offline datasets," *arXiv preprint arXiv:2006.09359*, 2020.

[21] J. Chen, Z. Xu, and M. Tomizuka, "Cooperative driving at unsignalized intersections using tree search," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 11, pp. 4563–4571, 2020.

[22] T. Chu, J. Wang, L. Codecà, and Z. Li, "Multi-agent deep reinforcement learning for large-scale traffic signal control," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 3, pp. 1086–1095, 2019.

[23] Z. Shou and X. Di, "Multi-agent coordination for on-ramp merging with deep reinforcement learning," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 9, pp. 14 711–14 721, 2022.

[24] C. Letter and L. Elefteriadou, "Efficient control of fully automated connected vehicles at freeway merge segments," *Transportation Research Part C: Emerging Technologies*, vol. 80, pp. 190–205, 2017.

[25] M. Zhou, Y. Yu, and X. Qu, "Multi-agent reinforcement learning for cooperative lane changing of connected and autonomous vehicles in mixed traffic," *Autonomous Intelligent Systems*, vol. 2, no. 1, pp. 1–14, 2022.

[26] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of ppo in cooperative multi-agent games," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 611–24 624, 2022.

[27] A. Haydari and Y. Yilmaz, "Deep reinforcement learning for intelligent transportation systems: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 1, pp. 11–32, 2020.

[28] L. Li, Y. Lv, and F.-Y. Wang, "Reinforcement learning for autonomous driving: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 6, pp. 5921–5940, 2023.

[29] Federal Highway Administration, "NGSIM – next generation simulation," U.S. Department of Transportation, Tech. Rep., 2006, dataset: Interstate 80 Freeway, Emeryville, CA.

[30] J. Colyar and J. Halkias, "US highway 101 dataset," in *Federal Highway Administration (FHWA), Tech. Rep. FHWA-HRT-07-030*, 2007.

[31] C. Thiemann, M. Treiber, and A. Kesting, "Estimating acceleration and lane-changing dynamics from next generation simulation trajectory data," *Transportation Research Record*, vol. 2088, no. 1, pp. 90–101, 2008.

[32] M. Montanino and V. Punzo, "Trajectory data reconstruction and simulation-based validation against macroscopic traffic patterns," *Transportation Research Part B: Methodological*, vol. 80, pp. 82–106, 2015.

[33] Z. Wu, Z. Sun, K. Zhang, and L. Chen, "Human-like autonomous driving: A survey," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 6, pp. 3538–3557, 2023.

[34] F. A. Oliehoek and C. Amato, "A concise introduction to decentralized pomdps," *Springer Briefs in Intelligent Systems*, 2016.

[35] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, "Microscopic traffic simulation using SUMO," in *21st International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2018, pp. 2575–2582.

[36] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. John Wiley & Sons, 2006.

[37] P. Christodoulou, "Soft actor-critic for discrete action settings," *arXiv preprint arXiv:1910.07207*, 2019.

[38] M. Jeragh and E. Alrashed, "Multi-agent reinforcement learning for highway exit coordination in connected vehicles," *Journal of Intelligent Transportation Systems*, 2026, under review, Manuscript #269158218.